

The Chang School

of Continuing

Education

Fall 2026

CIND 820 D1H – Big Data Analytics Project - F2026

Milestone 2: Architecture and Data Audit

Identifying Ontario School Boards Below the Provincial Five Years Graduation Benchmark Using Secondary-School Progress Indicators

October 19, 2026

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Abbreviations

HEQCO: Higher Education Quality Council of Ontario

OME: Ontario Ministry of Education

OSBAP: Ontario School Board Achievement and Progress

OSSLT: Ontario Secondary School Literacy Test

PII: Personally Identifiable Information

TDSB: Toronto District School Board

I. METHODOLOGICAL RATIONALE

1. Competitive Comparison of Approaches

High school graduation is an important indicator of educational success, but graduation is the result of a process that develops over several years. For education decision makers, a key challenge is therefore to determine whether information available before graduation can provide useful signals about later completion outcomes. Existing research provides strong support for examining this situation. Studies conducted in Ontario and internationally have repeatedly shown that credit accumulation, course performance, and literacy achievement are associated with eventual high school completion.

One of the most important Ontario studies is the Double Cohort Study (King, Warren, Boyer, and Chin, 2005). The study examined student progress following major reforms to Ontario secondary education and paid particular attention to credit accumulation and graduation. King and colleagues concluded that *"Credit loss in required courses is the biggest factor affecting graduation rates*."[1] Their analysis showed that students who failed to accumulate sufficient credits during the early years of secondary school were less likely to remain on a direct path toward graduation. The study also demonstrated that credit accumulation was already meaningful by Grade 10. Students progressing successfully tended to accumulate more credits than students experiencing academic difficulties. This is particularly relevant to the present project because Ontario continues to report the percentage of students completing at least 16 credits by the end of Grade 10 and at least 23 credits by the end of Grade 11.

The longitudinal research (Robert S. Brown, 2008) for the Toronto District School Board provides additional Ontario evidence. Brown followed Grade 9 student cohorts through secondary school and examined credit accumulation, literacy-test performance, attendance, and graduation. His findings showed a strong relationship between early credit achievement and later completion. Brown observed that "*those who finished 16 credits were most likely to graduate from high school.*"[2]

[1] Alan J. C. King, W. K. Warren, J. C. Boyer, and P. Chin, *Double Cohort Study: Phase 4 Report* (Ontario Ministry of Education, 2005), Executive Summary, p. ii.

[2] Robert S. Brown, *The Grade 9 Cohort of Fall 2002: A Five-Year Cohort Study, 2002–2007* (Toronto District School Board, 2008), p. 10.

Together, the work King and Brown supports the use of Grade 10 and Grade 11 credit accumulation as meaningful indicators of student progress. However, these studies examined individual students or cohorts over time. The current project differs because it examines aggregated indicators at the school board level.

Literacy achievement may also provide useful information about secondary school progress. Brown's TDSB (Toronto District School Board) research found that students who successfully completed the Ontario Secondary School Literacy Test (OSSLT) were generally more likely to graduate than students who initially failed, were absent, or were deferred. However, many students who initially did not succeed later completed the literacy requirement and graduated. This finding is important because it suggests that literacy achievement should be treated as an indicator, rather than as a deterministic measure of whether a student will graduate. International research has reached similar conclusions about the value of early academic indicators. Allensworth and Easton (Allensworth and Easton, 2007) studied Chicago Public Schools and examined course grades, failures, attendance, and credit accumulation during the first year of high school. They concluded that "*data on course performance can be used to identify future dropouts and graduates with precision.*"[3] Their research showed that students classified as academically "on track" early in high school were substantially more likely to graduate than students classified as "off track". The importance of this work for the present project lies in its decision support perspective: educational indicators can help identify cases requiring additional attention before the final graduation outcome is observed. Balfanz, Herzog, and Mac Iver (Balfanz, Herzog, and Mac Iver , 2007) similarly developed an early warning approach based on measurable student indicators. Their research demonstrated that "academic performance, attendance, and behavioural indicators could help identify students at greater risk of not completing high school."[4]

These studies support the broader principle that existing educational data can be used for screening and early identification, rather than waiting until graduation outcomes are already known. More recent Ontario research by Robson, Maier, Anisef, and Brown (Robson, Maier, Anisef, and Brown, 2019) also supports the relationship between academic progress and graduation. Their study for the Higher Education

[3] Elaine M. Allensworth and John Q. Easton, *What Matters for Staying On-Track and Graduating in Chicago Public High Schools* (Chicago: Consortium on Chicago School Research, 2007), p. 2.

[4] Robert Balfanz, Liza Herzog, and Douglas J. Mac Iver, "Preventing Student Disengagement and Keeping Students on the Graduation Path in Urban Middle-Grades Schools," *Educational Psychologist* 42, no. 4 (2007): 223–235.

Quality Council of Ontario (HEQCO) examined factors associated with high school graduation and access to post-secondary education using longitudinal Toronto student data. Reviewing previous Ontario evidence, the authors noted that "*students who fell below standard rates of credit accumulation were less likely to graduate and to attend PSE.*"[5] Their analysis considered academic achievement, credit accumulation, literacy test performance, absenteeism, program pathways, and student characteristics. The study is particularly useful for the present project because it also demonstrates an important limitation: graduation cannot be fully explained by only one or two academic indicators. Attendance, socioeconomic conditions, program pathways, special education, school context, and other factors may also influence student outcomes. Therefore, Grade 10 literacy and credit accumulation should not be interpreted as complete explanations of graduation performance. They are better understood as observable progress indicators that may contain useful information about broader graduation outcomes.

The literature provides strong evidence that academic progress indicators are associated with secondary school completion. Ontario studies by King and Brown emphasize the importance of credit accumulation, while Robson and colleagues confirm that academic progress remains an important component of graduation analysis. International research by Allensworth and Easton and by Balfanz and colleagues further supports the use of early indicators as screening tools.

However, most previous research has examined individual students or longitudinal student cohorts. The current project has a different and narrower objective. It examines whether Grade 10 literacy, Grade 10 credit accumulation, and Grade 11 credit accumulation are associated with five years graduation outcomes at the Ontario school board level. The project therefore does not attempt to predict whether individual students will graduate or to establish causal relationships. Instead, it investigates whether publicly available board level indicators can help distinguish school boards whose five years graduation rates fall below the provincial benchmark. This represents a practical decision support application of findings established in the literature. If the selected indicators can reliably distinguish below benchmark boards, they may provide a simple screening mechanism for identifying boards that warrant

[5] Karen Robson, Reana Maier, Paul Anisef, and Robert S. Brown, *High School Success and Access to Postsecondary Education* (Toronto: Higher Education Quality Council of Ontario, 2019), p. 12.

closer review. If they cannot, the project will demonstrate that additional contextual information is required.

2. Baseline Identification

A majority class classifier will serve as the primary baseline for evaluating the machine learning component. Among the approximately 70 school boards with complete secondary school data, 40 fall below the provincial five years graduation benchmark and 30 are at or above it. A baseline that classifies every board as below the benchmark would therefore achieve an accuracy of approximately 57.1%. This baseline represents a no information prediction because it does not use Grade 10 literacy, Grade 10 credit accumulation, or Grade 11 credit accumulation. The performance of the proposed logistic regression and decision tree models will be compared against this baseline to determine whether the secondary school progress indicators provide additional screening value. Evaluation will consider not only accuracy but also precision, recall, F1-score, specificity, and the confusion matrix, since a majority class model can achieve moderate accuracy while failing to distinguish between the two groups. A secondary comparison may use a model based on a single progress indicator, such as Grade 11 credit accumulation, to determine whether combining multiple indicators provides meaningful improvement. The machine learning approach will be considered useful only if it demonstrates improved and reasonably stable classification performance over the baseline while remaining interpretable for education decision makers.

3. Gap Analysis and Justification

a. Gap Analysis

Research consistently shows that secondary school completion is associated with academic progress, school engagement, and earlier indicators of student difficulty. In Ontario, Robson, Malette, Anisef, Maier, and Brown examined two Toronto District School Board cohorts and found that graduation was associated with academic achievement, absenteeism, program pathway, and student background characteristics. They also noted that "*additional data infrastructure in Ontario and beyond are necessary to determine how well such findings generalize provincially*."[6]

[6] Karen Robson, Nicole S. Malette, Paul Anisef, Reana Maier, and Robert S. Brown, "A Comparison of High School Graduation Predictors Between Two Ontario Student Cohorts," *Canadian Journal of Education* 45, no. 4 (2022): 1030.

International research similarly supports the use of observable indicators to identify graduation risk. Finn and Rock (Finn and Rock, 1997) found significant differences in school engagement behaviours between successful completers, lower performing completers, and students who dropped out. Balfanz, Herzog, and Mac Iver (Balfanz, Herzog, and Mac Iver, 2007) demonstrated that attendance, behaviour, and course failure indicators could identify a substantial proportion of students who would later fail to graduate. Their longitudinal study followed nearly 13,000 students and showed that a small set of indicators could identify approximately 60% of eventual non graduates. Other research emphasizes that dropout develops over time rather than resulting from a single event. Alexander, Entwisle, and Kabbani (Alexander, Entwisle, and Kabbani, 2001) describe dropout as the culmination of a long term disengagement process involving academic, family, and personal resources. Battin-Pearson and colleagues (Battin-Pearson and al., 2001) likewise found that poor academic achievement played an important role in dropout risk, while socioeconomic conditions, peer influences, and behavioural factors also contributed. These findings caution against interpreting any single academic indicator as a complete explanation of graduation outcomes. Canadian evidence supports the same multidimensional view. Archambault, Janosz, Fallu, and Pagani (Archambault, Janosz, Fallu, and Pagani, 2009) using a longitudinal sample of 11,827 French Canadian secondary school students, found that overall school engagement predicted dropout and that behavioural engagement made a particularly important contribution. A broader meta analysis by Gubbels, van der Put and Assink (Gubbels, van der Put, Assink, 2019) reviewed 33 dropout studies involving more than 136,000 students and found that low academic achievement and learning difficulties were among the stronger risk domains associated with school dropout.

The literature therefore provides substantial evidence that academic progress indicators can be useful for understanding graduation risk. However, most studies operate at the individual student or longitudinal cohort level. The proposed project addresses a different level of analysis: Ontario school boards.

The Ontario Ministry of Education currently publishes board level Grade 10 literacy, Grade 10 credit accumulation, Grade 11 credit accumulation, and five years graduation outcomes. Yet the literature reviewed above does not directly establish whether these publicly available indicators, when aggregated at the school board level, can distinguish boards whose five years graduation rates fall below the provincial benchmark. This creates a specific and manageable research gap. The

project will not attempt to reproduce student level causal or longitudinal research. Instead, it will test whether established secondary school progress indicators retain useful screening value at the board level.

b. Project Justification

The project is justified for three reasons. First, Ontario already collects and publishes these indicators, making it possible to evaluate their combined decision support value without collecting new student level data. Second, a simple screening approach could help provincial education decision makers identify boards that may warrant closer investigation rather than relying on the graduation rate alone. Third, the project provides a cautious test of whether simple and interpretable machine learning methods add useful information beyond a basic benchmark. The contribution is therefore practical rather than causal: to determine whether existing Ontario secondary school progress indicators can provide an interpretable board level screening signal for below benchmark five years graduation outcomes. A positive result would support further investigation of this approach, while a weak result would demonstrate that these indicators alone are insufficient and that additional contextual information is required.

4. 6+ Scholarly Sources

Alexander, K. L., Entwisle, D. R., & Kabbani, N. S. (2001). The dropout process in life course perspective. *Teachers College Record, 103*(5), 760–822.

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Archambault, I., Janosz, M., Fallu, J.-S., & Pagani, L. S. (2009). Student engagement and its relationship with early high school dropout**.** *Journal of Adolescence, 32*(3), 651–670.

Balfanz, R., Herzog, L., & Mac Iver, D. J. (2007). Preventing student disengagement and keeping students on the graduation path in urban middle-grades schools. *Educational Psychologist, 42*(4), 223–235.

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Brown, R. S. (2008). The Grade 9 cohort of fall 2002: A five-year cohort study, 2002–2007**.** Toronto District School Board.

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King, A. J. C., Warren, W. K., Boyer, J. C., & Chin, P. (2005). *Double cohort study: Phase 4 report*. Ontario Ministry of Education.

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Robson, K., Malette, N. S., Anisef, P., Maier, R., & Brown, R. S. (2022). A comparison of high school graduation predictors between two Ontario student cohorts. *Canadian Journal of Education, 45*(4), 1028–1054.

Ontario Ministry of Education. (2026). School board achievements and progress. Ontario Data Catalogue.

II. DATA PROFILING AND AUDIT

1. Univariate and Bivariate Analysis

a. Data Preparation

	Grade 10 OSSLT	Grade10 Credits	Grade 11 Credits	Five Years Graduuation
Min	0.570 000	0.630 000	0.650 000	0.690 000
Max	0.970 000	0.950 000	0.950 000	0.974 000
Mean	0.840 000	0.807 000	0.831 000	0.875 329
Median	0.845 000	0.810 000	0.845 000	0.883 000
Std	0.067 566	0.075 000	0.070 878	0.061 000

Data preparation resulted in a clean analytical dataset containing 70 Ontario school boards with complete secondary school performance information. Provincial aggregate rows were excluded from the modelling sample but were used to establish the provincial five years graduation benchmark of 89.5%. Four school boards with missing secondary school indicators were removed using complete case analysis rather than imputation because the amount of missing information was small.

The final dataset retained Grade 10 OSSLT results, Grade 10 credit accumulation, Grade 11 credit accumulation, and the five years graduation rate as the principal analytical variables. All selected measures were confirmed as numeric percentages,

no duplicate board records were identified, and no missing values remained in the modelling variables.

The distributions of the selected numerical variables were also examined using histograms and skewness measures. The observed skewness was predominantly negative (left-skewed), reflecting the concentration of many school boards toward the higher end of the performance scale. A conventional logarithmic transformation was therefore not applied, since a direct log transformation is primarily useful for positively or right skewed distributions and could increase rather than reduce negative skewness. In addition, normality of the predictor variables is not a requirement for the planned logistic regression analysis. The original percentage values were therefore retained to preserve their interpretability.

A binary classification target was created to identify whether each board's five years graduation rate was below the provincial benchmark. Of the 70 usable boards, 40 (57.1%) were below the benchmark and 30 (42.9%) were at or above it. This produces a reasonably balanced classification outcome and avoids the need for class balancing techniques at the data preparation stage.

Overall, the cleaned dataset is sufficiently complete, consistent, and interpretable for descriptive statistical profiling, correlation and bivariate analysis, and subsequent simple machine learning modelling. Retaining the variables in their original percentage scale also facilitates direct interpretation of the statistical and modelling results by education decision makers.

b. Statistical Profiling: Univariate and Bivariate Analysis

For this project, the statistical profiling should remain focused on the variables directly linked to the three research questions. After excluding the provincial aggregate rows and the two boards with incomplete secondary-school data, the analytical sample contains 70 Ontario school boards.

The four main numeric variables are:

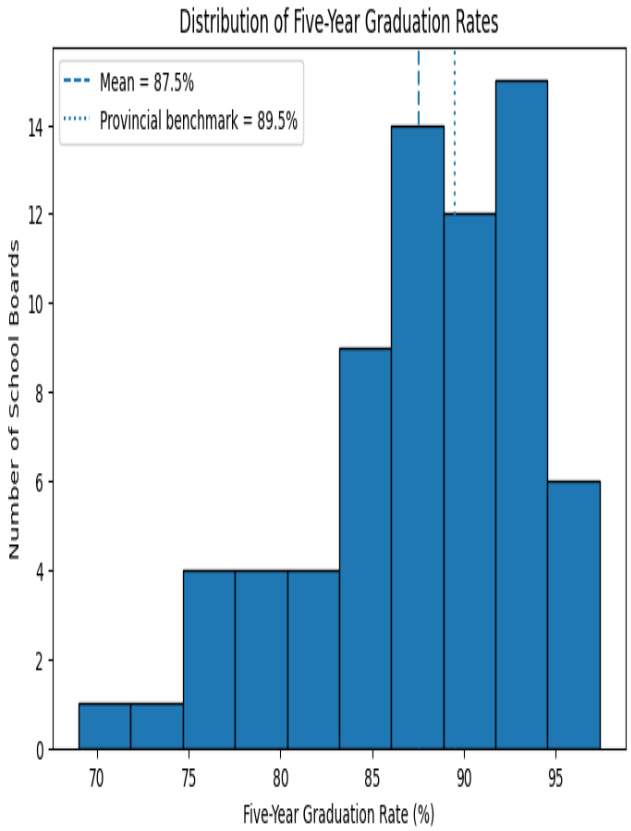
- Grade 10 OSSLT result
- Grade 10 credit accumulation
- Grade 11 credit accumulation
- Five-year graduation rate

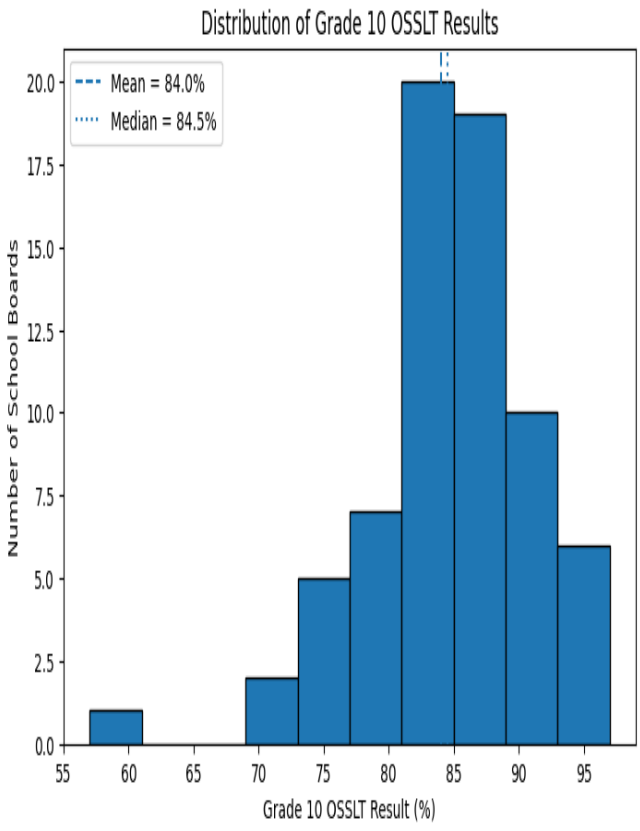
All percentages can be analyzed as ordinary numeric variables; for presentation, values such as 0.895 should simply be displayed as 89.5%.

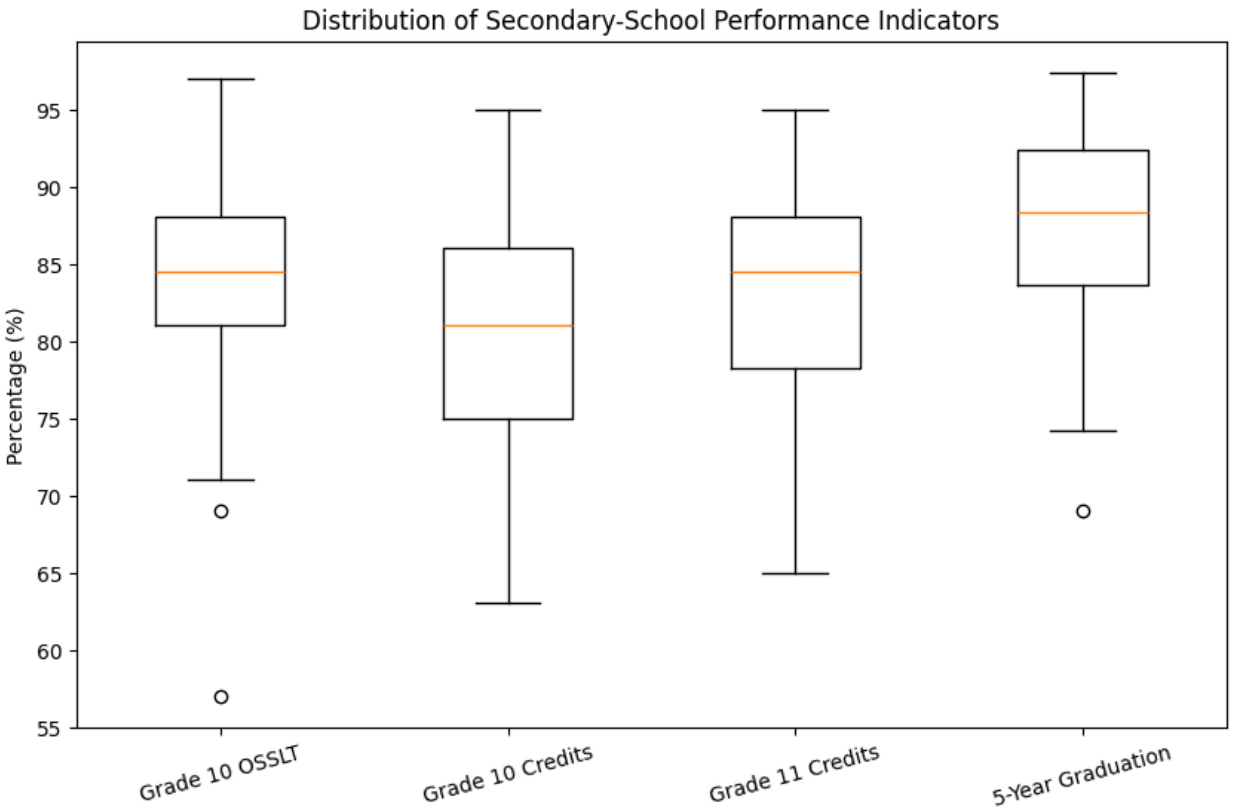
i. Univariate Analysis

The first stage examines each variable independently to understand its distribution, central tendency, variability, and potential outliers. The results show enough variation across boards to support further statistical analysis. Grade 10 credit accumulation shows the greatest standard deviation among the three progress indicators, while five-year graduation rates are somewhat less dispersed.

The distributions of Grade 10 OSSLT and five-year graduation rates show some negative skewness, meaning most boards are concentrated toward the higher end of the scale, with a smaller number of boards having substantially lower results. Credit accumulation variables are closer to symmetric. Using the provincial five years graduation benchmark of 89.5%: 40/70=57.1% of usable boards fall below the benchmark, while 30/70=42.9% are at or above it. This is particularly useful because the two groups are not extremely imbalanced.







The boxplots show the distribution and variability of the four main numeric variables: Grade 10 OSSLT results, Grade 10 credit accumulation, Grade 11 credit accumulation, and five years graduation rates. Most school boards are concentrated toward the higher percentage ranges, while a smaller number of boards have noticeably lower values. The medians are relatively high across all four indicators, but the spread differs between variables. The plots also suggest some lower end outliers, particularly for Grade 10 OSSLT and graduation outcomes. Overall, the distributions indicate meaningful variation across Ontario school boards, which supports further bivariate and modelling analysis.

	Frequency	**Percentage**
At or Above Benchmark	30	42.9
Below Benchmark	40	57.1

The frequency table shows that 40 of the 70 school boards, or 57.1%, have five years graduation rates below the provincial benchmark of 89.5%, while 30 boards, or 42.9%, are at or above the benchmark. The two classes are therefore reasonably balanced, although below benchmark boards form the larger group. This distribution is suitable

for classification analysis and provides a majority class baseline accuracy of approximately 57.1%.

Together, the boxplots and frequency table show that the dataset contains sufficient variation in both the continuous indicators and the binary graduation outcome to justify further correlation analysis and simple classification modelling.

ii. Bivariate Analysis: Progress Indicators Versus Graduation Rate

The second stage examines relationships between each progress indicator and the continuous five years graduation rate.

The Pearson correlations are:

Relationship	Pearson r	Interpretation
Grade 10 OSSLT - 5-years graduation	**0.745**	**Strong positive**
Grade 10 credits - 5-years graduation	**0.728**	**Strong positive**
Grade 11 credits ↔ 5-year graduation	**0.845**	**Very strong positive**

All three variables therefore show clear positive relationships with five years graduation outcomes. The strongest relationship is between Grade 11 credit accumulation and the five years graduation rate. In practical terms, boards with higher Grade 11 credit accumulation also tend to have higher five years graduation rates. This supports the rationale for retaining Grade 11 credit accumulation as an important predictor in the later modelling stage. These results demonstrate board level association only.

Predictor Pair	Pearson r
OSSLT - Grade 10 credits	0.647
OSSLT - Grade 11 credits	0.713
Grade 10 credits - Grade 11 credits	0.838

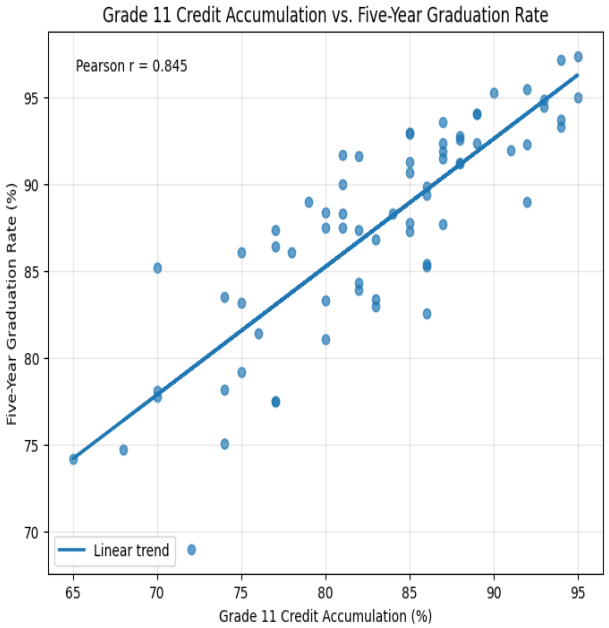
The predictors themselves are also correlated:

The correlation of 0.838 between Grade 10 and Grade 11 credit accumulation is important. It suggests that the two variables contain substantially overlapping information. This does not mean one must automatically be removed, but it does mean that multicollinearity should be checked before interpreting a regression or logistic regression model. This is exactly the type of finding statistical profiling is supposed to reveal before modelling.

Indicators	Below 89.5%	At/Above 89.5%
Mean Grade 10 OSSLT	**81.2%**	**87.7%**
Mean Grade 10 credit accumulation	**77.6%**	**84.9%**
Mean Grade 11 credit accumulation	**78.9%**	**88.7%**

For Research Question 3, it is also useful to compare the three indicators between boards below and at/above the provincial graduation benchmark. The largest difference appears in Grade 11 credit accumulation, with approximately 9.8 percentage points separating the two groups. This preliminary finding is important for the later machine learning analysis because it suggests that the selected indicators contain information capable of distinguishing the two graduation outcome groups.

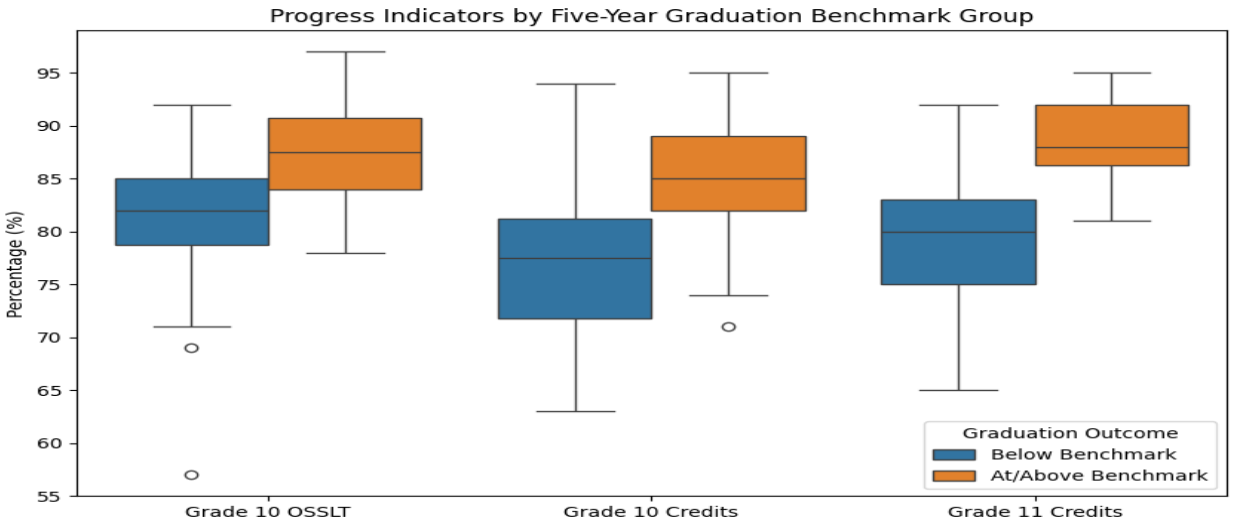
The scatterplot shows a clear positive relationship between Grade 11 credit accumulation and fives years graduation rates. School boards with higher percentages of students accumulating the expected number of Grade 11 credits generally also have higher five years graduation rates. The Pearson correlation is approximately r = 0.845, indicating a strong positive linear association. Grade 11 credit accumulation therefore appears to be an important indicator of board level graduation outcomes. However, this relationship represents an association and should not be interpreted as evidence that higher credit accumulation directly causes higher graduation rates.



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	Grade10 OSSLT	Grade10 Credits	Grade11 Credits	Five Years Graduation
Grade10 OSSLT	1.000000	0.647070	0.712996	0.744999
Grade10 Credits	0.647070	1.000000	0.837813	0.727537
Grade11 Credits	0.712996	0.837813	1.000000	0.845239
Five Years Graduation	0.744999	0.727537	0.845239	1.000000

The correlation matrix indicates positive relationships among all three secondary schools progress indicators and the five years graduation rate. Grade 11 credit accumulation has the strongest relationship with five years graduation (r ~ 0.845), followed by Grade 10 OSSLT achievement (r ~ 0.745) and Grade 10 credit accumulation (r ~ 0.728). These results suggest that boards performing better on the selected progress indicators also tend to report higher graduation rates. A strong relationship is also observed between Grade 10 and Grade 11 credit accumulation (r ~ 0.838). This indicates that the two predictors contain overlapping information and suggests that multicollinearity should be assessed before interpreting the coefficients of a multivariable model. The correlations demonstrate association rather than causation.



The grouped boxplots show clear differences in the secondary school progress indicators between boards below the provincial five years graduation benchmark and those at or above it. Boards at or above the benchmark generally have higher Grade 10 OSSLT results, Grade 10 credit accumulation, and Grade 11 credit accumulation. The difference appears especially pronounced for Grade 11 credit accumulation. On average, Grade 11 credit accumulation is approximately 78.9% among below benchmark boards compared with 88.7% among boards at or above the benchmark, a difference of approximately 9.8 percentage points. Similar, although smaller, differences are observed for Grade 10 OSSLT and Grade 10 credit accumulation. The distributions still overlap, indicating that no individual progress indicator perfectly separates the two groups. This supports evaluating the indicators together using a simple classification model rather than relying on a single variable.

These three graphs tell a coherent story: the scatterplot establishes that Grade 11 credit accumulation is strongly associated with graduation. The correlation matrix establishes that all three predictors are related to graduation, while also detecting possible multicollinearity between Grade 10 and Grade 11 credits. The grouped boxplots establish that the predictor distributions differ between the two classes we ultimately want to classify. Together, they provide a strong statistical justification for moving from CIND123 exploratory analysis to the logistic regression stage.

2. Data Structure and Integration Plan

The Ontario School Board Achievements and Progress (OSBAP) dataset is a structured, tabular dataset provided in csv format. Each row represents a school board level observation, while columns contain clearly defined variables such as board characteristics, Grade 10 OSSLT results, Grade 10 and Grade 11 credit accumulation, and graduation outcomes. Because the data are already organized into rows and columns, no parsing or flattening of semi-structured data is required. Preparation therefore consisted of cleaning column names, selecting relevant variables, converting analytical measures to appropriate numeric formats, identifying missing values, removing provincial aggregate records, and creating a binary outcome indicating whether a board's five years graduation rate was below the provincial benchmark. The dataset is not sufficiently large to require construction of a representative sub population. Instead, all eligible school boards with complete secondary school information were retained. Provincial aggregate records were excluded because they are summary observations rather than individual boards, and

records with incomplete values for the required analytical variables were removed through complete case analysis. The resulting analytical dataset contains 70 Ontario school boards. Using the full eligible population avoids unnecessary sampling error and preserves as much information as possible for the subsequent statistical and machine learning analyses.

No multi source data integration is required for this project because the variables needed to answer the research questions are already available within a single Ontario Ministry of Education (OME) dataset. The final working dataset is therefore constructed by selecting the relevant school board level variables, removing aggregate records, handling missing values, validating numeric fields, and creating the derived below benchmark classification target. If an additional dataset were incorporated, board number would be used as the primary integration key because it provides a consistent identifier across school boards. A Pandas merge () would be the preferred integration method, followed by validation of matched and unmatched records, duplicate keys, missing values, and row counts. Since no additional source is currently required, avoiding unnecessary integration keeps the analytical pipeline simpler and reduces the risk of data matching errors.

3. Constraints And Limitations

Several limitations affect the interpretation of this analysis. First, the final dataset contains only 70 school boards, which limits the use of complex machine learning methods and increases uncertainty in model performance estimates. The project will therefore rely on simple, interpretable models and cross validation. Second, the data are aggregated at the school board level. Consequently, relationships identified between literacy, credit accumulation, and graduation outcomes cannot be generalized to individual students or interpreted as student level effects. A particularly important limitation concerns time and cohort alignment. Grade 10 literacy and credit accumulation indicators and the five years graduation outcome do not represent the same student cohort and reporting period. The analysis therefore cannot claim that the Grade 10 or Grade 11 measures directly predict the later graduation of those same students. Instead, the results will be interpreted as cross-sectional associations between board level progress indicators and board level graduation outcomes. Two boards have incomplete secondary school data and were excluded through complete case analysis. Although the amount of missing data is small, the results cannot be assumed to represent the excluded boards. In addition,

the three selected predictors capture only part of the factors potentially associated with graduation. Attendance, socioeconomic circumstances, student demographics, school climate, local programs, and other contextual factors are not included. The model should therefore be considered a screening tool rather than a complete explanation of graduation performance. The strong correlation between Grade 10 and Grade 11 credit accumulation also creates a potential multicollinearity issue, which will be assessed during modelling. Finally, the 89.5% provincial graduation benchmark is used as an analytical threshold rather than as a judgement of board quality. Because the dataset is observational, all relationships will be reported as associations rather than causal effects, and conclusions will be limited to Ontario school boards and the reporting periods represented by the available data.

4. Privacy Check

The dataset contains no Personally Identifiable Information (PII). Its unit of analysis is the Ontario school boards, and all educational outcomes are reported as aggregated percentages rather than individual student records. No names, student identification numbers, addresses, dates of birth, information, or other individual level identifiers are included. Although the measures are derived from student educational data, the project uses only publicly released, aggregated board level information and makes no attempt to identify or reconstruct contact information about individual students. The present project uses an official public Ontario government dataset, involves no direct contact with human participants, and does not perform data linkage that could create identifiable information. Therefore, the project presents minimal privacy risk. Institutional research ethics requirements will nevertheless be followed if a formal determination is required.

III. ARCHITECTURE AND LOGIC

1. Sequence of Operations

The project will follow a clear sequence so that each stage builds on the previous one and the analysis remains reproducible.

a. Load the source dataset

Import the Ontario School Board Achievements and Progress CSV file into Python using Pandas.

b. Inspect the raw data

Review the number of rows and columns, variable names, data types, missing values, and unusual records.

c. Clean column names and data types

Remove extra spaces and encoding issues, rename key variables, and convert the analytical measures to numeric format.

d. Identify and remove non analytical records

Remove provincial aggregate rows and any other non school board summary records from the modelling dataset.

e. Handle missing values

Identify incomplete records and apply complete-case analysis for boards missing required secondary-school indicators.

f. Select the analytical variables

Retain:

- Grade 10 OSSLT results
- Grade 10 credit accumulation
- Grade 11 credit accumulation
- Five-year graduation rate

Relevant board identifiers and characteristics will also be retained for interpretation.

g. Validate data quality

Check for duplicate board records, invalid percentage ranges, remaining missing values, and inconsistent categorical values.

h. Create the classification target

Use the provincial five-year graduation benchmark of **89.5%** to create:

- 1 = Below benchmark
- 0 = At or above benchmark

i. Conduct univariate analysis

Examine distributions using: descriptive statistics; histograms; boxplots; skewness; frequency tables.

j. Conduct bivariate analysis

Examine relationships using: scatterplots; Pearson correlation; correlation; matrix/heatmap; grouped boxplots by benchmark class.

k. Assess modelling assumptions and predictor redundancy

Review strong correlations among predictors, particularly Grade 10 and Grade 11 credit accumulation, and assess potential multicollinearity.

l. Establish the baseline model

Use the majority-class classifier as the primary baseline, with expected accuracy of approximately 57.1%.

m. Build interpretable classification models

Train: logistic regression as the primary model; a shallow decision tree as a comparison model.

n. Evaluate model performance

Use stratified cross-validation and report: accuracy; precision; recall; F1-score; confusion matrix; false positives and false negatives.

o. Compare models with the baseline

Determine whether the progress indicators improve classification performance beyond the simple majority-class rule.

p. Interpret results and limitations

Explain which indicators contribute most to classification while avoiding causal claims and acknowledging the small sample size, board-level aggregation, and cohort-period mismatch.

q. Produce final outputs

Prepare tables, visualizations, model results, and stakeholder oriented conclusions.

r. Save the cleaned analytical dataset and reproducible code

Preserve the cleaned dataset and final notebook so that the complete workflow can be rerun from the original source data.

2. Preprocessing and Feature Engineering

a. Preprocessing

Preprocessing will convert the raw Ontario School Board Achievements and Progress (OSBAP) dataset into a reliable analysis ready table through a connected sequence of validation missing data treatment - outlier assessment - distribution assessment - target creation. This order is important because later statistical decisions should only be made after the underlying records have been verified. Van den Broeck and al. (Van den Breck and al., 2005) describe data cleaning as a "*three-stage process, involving repeated cycles of screening, diagnosing, and editing"* [7] suspected abnormalities.

First, the dataset will be checked for its unit of analysis, duplicates, invalid values, data types, and non board aggregate records. The provincial aggregate will be used to obtain the five years graduation benchmark of 89.5% and then removed from the modelling population. The selected educational indicators will be converted to

[7] Van den Broeck, J., Argeseanu Cunningham, S., Eeckels, R., & Herbst, K. (2005). "Data Cleaning: Detecting, Diagnosing, and Editing Data Abnormalities." *PLoS Medicine, 2*(10), e267, p. 967.

numeric values and checked to confirm that percentages fall within their valid range. This validation must precede statistical profiling because invalid records could distort distributions, correlations, and subsequent modelling.

Second, missingness will be examined before records are removed. Two of the 72 eligible school boards have incomplete secondary school indicators, representing approximately 2.8% of the board population. These two records will be excluded through complete case analysis rather than imputed. This decision is appropriate because the proportion of incomplete records is very small and imputing artificial performance percentages would add unnecessary assumptions. Lee and al. (Lee and al., 2021) note that *"if 5% of records are expected to have missing values, it will not make much difference how missing data are handled."*[8] The final analytical sample therefore contains 70 complete school boards. Nevertheless, the excluded boards will be identified and the missingness reported as a limitation rather than assumed to be random.

Third, potential outliers will be examined using descriptive statistics, histograms, and boxplots. Outlying values will not be deleted automatically, because an unusually low board result may represent genuine educational variation rather than an error. Van den Broeck and al. (Van den Broeck and al., 2005) emphasize that suspected observations may be diagnosed as *"erroneous, true extreme, true normal"*[9] or remain unexplained. Consequently, a value will only be corrected or excluded when there is evidence of an invalid or erroneous record; valid extreme observations will remain in the analysis.

Finally, skewness will be assessed before deciding whether transformation is necessary. The principal indicators show predominantly negative skewness, with many boards concentrated toward higher performance values. A logarithmic transformation will therefore not be applied automatically. Feng and al. (Feng and al., 2013) caution that *"in general, such a transformation does not reduce skewness"*[10] and may sometimes introduce additional skewness. Retaining the original percentage scale also preserves direct interpretation for education decision makers.

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[8] Lee, K. J., Tilling, K. M., Cornish, R. P., Little, R. J. A., Bell, M. L., Goetghebeur, E., Hogan, J. W., & Carpenter, J. R. (2021). "Framework for the Treatment and Reporting of Missing Data in Observational Studies." *Journal of Clinical Epidemiology, 134*, 79–88, p. 82.

[9] Van den Broeck, J., Argeseanu Cunningham, S., Eeckels, R., & Herbst, K. (2005). "Data Cleaning: Detecting, Diagnosing, and Editing Data Abnormalities." *PLoS Medicine, 2*(10), e267, p. 968.

[10] Feng, C., Wang, H., Lu, N., & Tu, X. M. (2013). "Log Transformation: Application and Interpretation in Biomedical Research." *Statistics in Medicine, 32*(2), 230–239, p. 233.

Only after these checks are complete will the binary target be created: 1 = below the 89.5% provincial benchmark and 0 = at or above the benchmark. Thus, every later statistical and modelling step depends on a validated, complete, and interpretable dataset.

b. Feature Engineering

Feature engineering will remain simple because the dataset already contains three interpretable numeric predictors: Grade 10 OSSLT, Grade 10 credit accumulation, and Grade 11 credit accumulation. No logarithmic transformation will be applied because the variables are predominantly negatively skewed, and logistic regression does not require normally distributed predictors (Bewick V and al., 2005).

For logistic regression, the three predictors will be standardized using StandardScaler. Scaling is appropriate because it places predictors on a common scale and supports stable estimation, particularly when regularization is used (Hastie T and al., 2009). To prevent data leakage, scaling will be performed inside the cross validation pipeline, so the mean and standard deviation are learned only from each training fold and then applied to its validation fold.

The decision tree will use the same three predictors without scaling, because tree based models are largely unaffected by feature scaling. This still provides a fair comparison because both models use the same observations, predictors, target variable, and cross validation folds.

Categorical board characteristics will not be included in the primary predictive model, so one hot encoding is not required. PCA will also not be used because there are only three predictors and preserving their direct interpretation is more important than dimensionality reduction (Hastie T. and al., 2009).

For fairness, logistic regression and the decision tree will therefore be compared using the same 70 boards, same three predictors, same binary target, same stratified cross-validation folds, and same evaluation metrics.

3. Modelling and Evaluation Strategy

The models will use Grade 10 OSSLT, Grade 10 credit accumulation, and Grade 11 credit accumulation as predictors. The output will be the probability that a board is below the 89.5% five years graduation benchmark, followed by a binary classification. This stage mainly answers recherche question 3, while recherche question 1 and recherche question 2 are addressed through descriptive and correlation analysis.

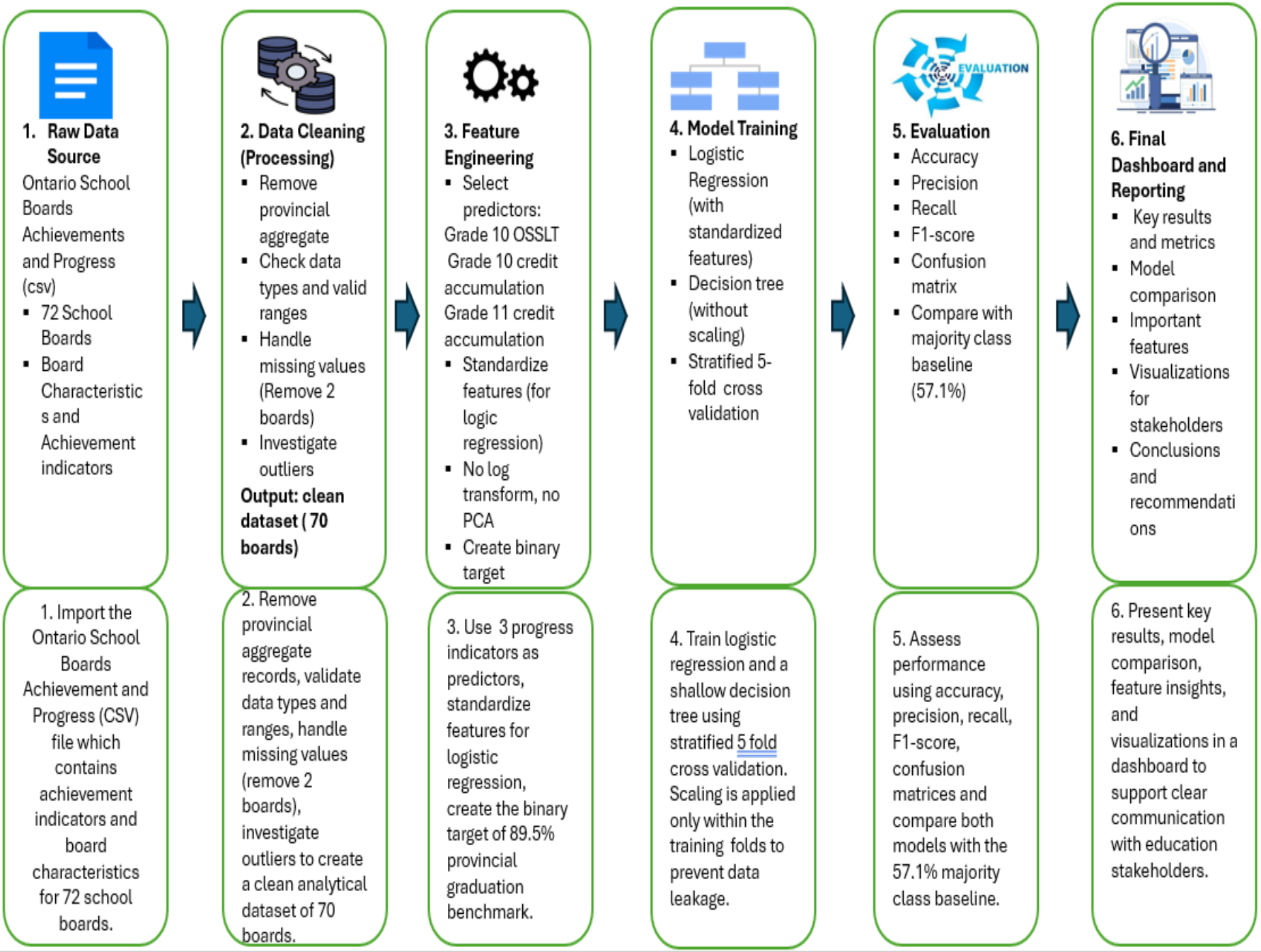
Two interpretable models will be compared: logistic regression and a shallow decision tree. Logistic regression assumes independent observations, limited multicollinearity, and an approximately linear relationship between continuous predictors and the log odds of the outcome (Bewick and al., 2005). The decision tree provides a useful comparison because it does not require linear relationships or normally distributed predictors. Because the dataset contains 70 cross sectional school boards, the evaluation will use stratified five folds cross validation. Stratification preserves the proportion of below and at/above benchmark boards in each fold and is appropriate for classification with unequal class frequencies (Kohavi R, 1995). Both models will use the same folds, predictors, and target to ensure a fair comparison.

To prevent leakage, StandardScaler for logistic regression will be fitted inside the cross validation pipeline, using only the training portion of each fold. Five years graduation rate will not be included as a predictor because it is used to create the target; four years graduation will also be excluded to avoid circular reasoning.

Performance will be assessed using accuracy, precision, recall, F1-score, and confusion matrices, and compared with the 57.1% majority class baseline. If logistic regression performs better, the relationships are likely to be relatively simple and additive; if the decision tree performs better, threshold or nonlinear effects may matter. If neither improves meaningfully on the baseline, the three indicators alone will be considered insufficient for screening.

4. Visual Pipeline diagram

The visual pipeline diagram can be seen here.


1. Raw Data Source
Ontario School Boards Achievements and Progress (csv)
72 School Boards
Board Characteristics and Achievement indicators
2. Data Cleaning (Processing)
Remove provincial aggregate
Check data types and valid ranges
Handle missing values (Remove 2 boards)
Investigate outliers
Output: clean dataset (70 boards)
3. Feature Engineering
Select predictors:
Grade 10 OSSLT
Grade 10 credit accumulation
Grade 11 credit accumulation
Standardize features (for logic regression)
No log transform, no PCA
Create binary target
4. Model Training
Logistic Regression (with standardized features)
Decision tree (without scaling)
Stratified 5-fold cross validation
EVALUATION
5. Evaluation
Accuracy
Precision
Recall
F1-score
Confusion matrix
Compare with majority class baseline (57.1%)
6. Final Dashboard and Reporting
Key results and metrics
Model comparison
Important features
Visualizations for stakeholders
Conclusions and recommendations
1. Import the Ontario School Boards Achievement and Progress (CSV) file which contains achievement indicators and board characteristics for 72 school boards.
2. Remove provincial aggregate records, validate data types and ranges, handle missing values (remove 2 boards), investigate outliers to create a clean analytical dataset of 70 boards.
3. Use 3 progress indicators as predictors, standardize features for logistic regression, create the binary target of 89.5% provincial graduation benchmark.
4. Train logistic regression and a shallow decision tree using stratified 5 fold cross validation. Scaling is applied only within the training folds to prevent data leakage.
5. Assess performance using accuracy, precision, recall, F1-score, confusion matrices and compare both models with the 57.1% majority class baseline.
6. Present key results, model comparison, feature insights, and visualizations in a dashboard to support clear communication with education stakeholders.

IV. REPRODUCIBILITY (GITHUB)

The GitHub repository can be found here.

V. INTERIM REPORT

1. Milestone 1 Updates

The first milestone covered the problem definition, decision context, beneficiaries, dataset selection and biography, research questions, scope, impact hypothesis, feasibility, ethics/privacy framing, and the overall analytical plan. The current milestone covered:

- data structure and preparation;
- data integration strategy;
- constraints analysis;
- privacy check;
- preprocessing;
- feature engineering;
- modelling and evaluation;
- system architecture diagram;
- GitHub repository structure and reproducibility.

2. Architectural Justifications

The system architecture was designed to remain proportional to the size and complexity of the data. Because the source is a small, structured csv dataset, Pandas and Google Colab are sufficient for ingestion, cleaning, analysis, and modelling; a database or distributed processing environment would add unnecessary complexity. Data cleaning is separated from feature engineering and modelling to ensure that only validated school board records enter the analytical pipeline. The three selected educational indicators are retained in interpretable form, while logistic regression and a shallow decision tree provide complementary modelling approaches. Stratified five folds cross validation is used because the sample contains only 70 boards, and all preprocessing required by a model is performed within the validation pipeline to prevent leakage. The final dashboard/reporting layer converts statistical and model outputs into transparent screening information for review rather than automated intervention.

3. AI Use Declaration

AI was not used to fabricate data, invent results, create unverified citations, or make final analytical decisions independently. All code was reviewed, executed, and validated using the actual Ontario school board dataset. Statistical results and model performance measures were generated from the project data. AI was only used to check the accuracy of the Python codes on Google Colab.

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